

Semester/Year:	III Sem / II Year
Course Code:	26MBAB001
Title Of the Course:	Data Management Systems
L: T: P: C:	3: 0: 2: 4

Overview

Today 's organizations recognize that managing data is central to their success. They recognize data has value and they want to leverage that value. Data is the fundamental raw material for any business which enable sensing the internal and external parameters that could have bearing on business performance. Those parameters provide the basis for action when analysed for insights. The insights are relevant to understand the current state of the business as well as the likely future state. The derived insights provide the basis for all business decisions which precede action. Having the required insights in time, enable the organisations to be responsive and adapt. Harnessing required data while ensuring its dependability organised storage to enable access on demand, data security and access control is the subject of data management systems. As all aspects of business have dependency on data, every business executive has a definite role in contributing to the success of data management systems. Business organizations are adapting to become digital enterprises of the future. Understanding data management systems facilitates active participation in transformation process to digital enterprise.

Course Objectives

1. To enable students to appreciate data requirements for business
2. To enable students to understand data and data management systems in relation to any business domain by a systematic process based on goal setting and performance management as the factors that lead to success
3. To enable understanding of data sources, data capture & storage
4. To enable students to understand data models, master data, MDM, schema, relational and No-SQL databases, data warehouse and data lake
5. To enable understanding of data organisation from speed of response, and information security

Course Outcomes

After completion of this course students will be able to:
1. Exhibit the fair understanding of concepts, theories, models and principles of business data management course, including data requirements, business entities, their relationships, and the business value of data.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of business data management in real-time problem solving by identifying and harnessing various data sources such as ERP systems, big data, databases, data warehouses, NoSQL databases, ETL systems, IoT, and data streams.
3. Demonstrate the ability of analysing real-time problems using the skills acquired from concepts, theories, models and principles of business data management by evaluating master data, organizational data dictionaries, and one-truth databases.

4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of business data management for decision-making related to data centres, including in-house, co-located, and cloud computing (private, public, hybrid).
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of business data management to formulate data security requirements and policies.

Approach to Learning	• Lectures • Readings • Active student participation and classroom exercises • Case analysis collaboratively with student involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of data management systems by way of: • Mini projects • Submission of assignments • Group assignments • Written exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	Data Fundamentals Data, types of data, information, derived insights from analysis, meta data, data structures, decision making in management - Strategic, tactical and operational decisions, decision support with management information systems, data sources & defining data need, internal data, transaction data, structured and unstructured data, external data, PESTEL, competition, Porters 5 forces, physical records, flat files, spreadsheet, databases, social media, data models, retail marketing as a case study to exemplify definition of data requirements, information security, data life cycle management, types of data models: Conceptual, logical, physical, case study on data governance in retail sector, introduction to data profiling techniques, key performance indicators (KPIs) and their data sources, real-world examples of MIS in decision-making, security risks in data handling and prevention measures.	8

Unit II	Database Systems Data base systems, relational databases (RDBMS) in particular, entity relations and ER diagram, database schema, data base organisation, primary key, and foreign key, stored procedures and referential integrity, data base replication, introduction to SQL for query and reporting with examples, popular RDBMS with relative merits, normalization: 1NF, 2NF, 3NF, BCNF, indexing for query optimization, Joins: Inner, outer, left, right, stored function vs stored procedure.	8
Unit III	Using SQL for Extracting Information from RDBMS Using SQL for extracting information from RDBMS, introduction to entity and entity relations, attributes, Entity – Primary and foreign keys, SQL commands, using MS-SQL express to work on creating, modifying, querying, reporting, aggregate functions: SUM, AVG, COUNT, GROUP BY and HAVING clauses, pattern matching with LIKE, SQL date and time functions.	8
Unit IV	Data Warehouse, Data Lakes and their Applications Data warehouse, data lakes and their applications, data warehouse and data mart, data warehouse schema, star, snowflake, Star - Operational data store, ETL, OLAP, limitations of data warehouse, data lake, concepts, business purpose, well known data warehouse and data lake products, data virtualization concepts, data ingestion tools (e.g., Apache NiFi, Talend), slowly changing dimensions (SCD), data governance in warehousing, integration of cloud storage with data lakes.	8
Unit V	Big Data, Unstructured Databases (NoSQL), and Data Mining Big data, unstructured databases aka NoSQL databases, data mining, big data, concepts and uses, introduction and the need for unstructured data bases, organization of NoSQL databases, top rated NoSQL databases, extracting information from NoSQL databases, introduction to data mining, business purpose served, data structures for data mining, CAP theorem in NoSQL, types of NoSQL databases: Key-value, document, column, graph, big data architecture overview, data preprocessing for data mining, ethical considerations in big data analytics.	8

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4
CO1	2	2	2	2	2	2	2	2	2
CO2	3	2	2	2	2	2	3	2	2
CO3	3	3	3	3	2	2	2	2	3
CO4	3	2	2	2	2	2	2	2	2
CO5	3	2	2	2	2	2	2	3	2

Action Based Component

1. Defining data requirements for finance, marketing & sales, HR, SCM and CRM
2. Use SQL query to extract information from MS-SQL express. Students will be installing MS-SQL express, load data and use SQL to query and extract data
3. Design a mini data warehouse
4. Work on NoSQL using Mongo-SQL or any other NoSQL Databases depending on free license availability
5. Mini workshop on information security needs, access control

Course Assessment

S.No.	Description of Assessment Method	Weightage %	Learning Outcomes Assessed				
			1	2	3	4	5
1	Assignment-1	15	X	X	X	X	X
2	Assignment- 2	15	X	X	X	X	X
3	Internal Assessment-1	15	X	X	X	X	X
4	Internal Assessment-2	15	X	X	X	X	X
5	Semester End Examination	40	X	X	X	X	X

Recommended Resources

Text Books
1. Connolly, T. M., & Begg, C., 2019. Database Systems: A Practical Approach to Design, Implementation, and Management, 6 th ed., Pearson Education.
2. Silberschatz, A., Korth, H. F., & Sudarshan, S., 2013. Database System Concepts, 6 th ed., McGraw Hill Education.
3. Pratt, P. J., & Adamski, J. J., 2012. Database Management Concepts, 7 th ed., Cengage India.
Reference Books
1. DAMA International, 2017. DAMA-DMBOK: Data management body of knowledge, 2 nd ed., Technics Publications LLC.
2. Chaffey, D., 2015. E-business and e-commerce management: Strategy, implementation and practice, 6 th ed., Pearson Education.
3. O'Brien, J. A., Marakas, G. M., & Behl, R., 2018. Management information systems, 11 th ed., McGraw Hill Education (India).
4. Berson, A., & Smith, S. J., 2015. Data warehousing, data mining, and OLAP, 3 rd ed., McGraw Hill Education.

5. DT Editorial Services, 2015. Big data: Big data black book: Covers Hadoop 2, MapReduce, Hive, YARN, Pig, R and data visualization, Dreamtech Press.
6. Kukreja, M., & Zburivsky, D., 2021. Data engineering with Apache Spark, Delta Lake, and Lakehouse: Create scalable pipelines that ingest, curate, and aggregate complex data in a timely and secure way, Packt Publishing.
7. Whitman, M. E., & Mattord, H. J., 2018. Principles of information security, 6 th ed., Cengage Learning.
8. Schwab, K., 2016. The fourth industrial revolution, Penguin Random House.

Further Readings

1. https://docs.oracle.com/database/121/TDPDW/tdpdw_arch.htm#TDPDW380
2. https://docs.informatica.com/data-integration/data-services/10-2/application-service-guide/content-management-service/reference-data-warehouse.html
3. https://www.bigdataframework.org/big-data-architecture/#:~:text=%20The%20NIST%20Big%20Data%20Reference%20Architecture%20,can%20be%20used%20by%20the%20Big...%20More%20
4. https://www.geeksforgeeks.org/data-architecture-design-and-data-management/
5. https://pubs.opengroup.org/onlinepubs/9199929899/toc.pdf
6. https://aws-reference-architectures.gitbook.io/datalake/

Suggested Courses

1. IBM Data Engineering Professional Certificate Coursera: https://www.coursera.org/professional-certificates/ibm-data-engineer
2. Data Warehousing for Business Intelligence Specialization Coursera: https://www.coursera.org/specializations/data-warehousing

Semester/Year:	III Sem / II Year
Course Code:	26MBAB002
Title of the Course:	Applied Analytics
L: T: P: C:	3: 0: 2: 4

Overview

The Applied Analytics course is designed not only to explain what each model does or functions but also explores how businesses use them, whether it is to gather insights, solve problems, or predict outcomes.

Course Objectives

1. To apply quantitative modelling and data analysis techniques to the solution of real-world business problems, communicate findings, and effectively present results
2. To demonstrate knowledge of statistical data analysis techniques utilized in business decision making
3. To apply principles of data science to the analysis of business problems
4. To use data mining software to solve real-world problems
5. To employ cutting edge tools and technologies to analyse (Big) data

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of business analytics course, including how to collect and prepare data for analysis
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of business analytics in real-time problem solving by using analytical tools to gather meaningful business insights
3. Demonstrate the ability of analyzing real-time problems using the skills acquired from concepts, theories, models and principles of business analytics through data-driven decision-making to support informed business decisions
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired

skills from concepts, theories, models and principles of business analytics to examine whether analytical results confirm business hypotheses
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of business analytics, including evaluating ways to apply machine learning techniques to solve business problems

Approach to Learning	• Lectures • Readings • Active student participation and classroom exercises • Case analysis collaboratively with student involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business applications of applied analytics by way of: • Mini projects • Submission of assignments • Group assignments • Written exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	Spreadsheet Modelling Spreadsheet modelling, concepts and best practices, relative & absolute addresses, range names, auditing formulas, SUM, AVERAGE, PRODUCT, COUNT, COUNTA, COUNTBLANK, COUNTIF, SUMIF, AVERAGEIF, COUNTIFS, SUMIFS, AVERAGEIFS, SUMPRODUCT, IF, VLOOKUP, RAND, RANDBETWEEN, pivot tables, data tables, goal seek, INDEX and MATCH functions, using TEXT functions (LEFT, RIGHT, MID, TEXT), date and time functions.	8
Unit II	Marketing Marketing, predicting customer churn, customer segmentation, sales forecasting, customer lifetime value analysis, A/B testing for marketing campaigns.	8
Unit III	Human Resource Resource, predictive HR analytics in recruitment and selection, predictive HR analytics in turnover and separation, predictive HR analytics in learning and development, attrition risk scoring models, absenteeism prediction using historical data.	8
Unit IV	Finance Finance, predicting stock market returns, detecting fraudulent transactions, predicting bank loan defaults, loan approval scoring systems, Value at Risk (VaR) estimation.	8
Unit V	Supply Chain Supply chain, Monte Carlo simulation modelling (Harry 's Auto Tire), transportation problem (Executive Furniture Corporation), waiting line problem (Three Rivers Shipping Company), inventory optimization using ABC analysis, safety stock calculation and reorder point models.	8

CO- PO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4
CO1	2	2	2	2	2	2	2	2	2
CO2	3	2	2	2	2	2	3	2	2
CO3	3	3	3	3	2	2	2	2	3
CO4	3	2	2	2	2	2	2	2	2
CO5	3	2	2	2	2	2	2	3	2

Action Based Component

1. Advanced excel functions and formulas using Microsoft excel 2019
2. Supervised machine learning XLMiner
3. Unsupervised machine learning XLMiner
4. Deterministic optimisation using Microsoft excel solver
5. Monte Carlo simulation using Microsoft excel 2019

Course Assessment

S.No.	Description of Assessment Method	Weightage %	Learning Outcomes Assessed				
			1	2	3	4	5
1	Assignment-1	15	X	X	X	X	X
2	Assignment- 2	15	X	X	X	X	X
3	Internal Assessment-1	15	X	X	X	X	X
4	Internal Assessment-2	15	X	X	X	X	X
5	Semester End Examination	40	X	X	X	X	X

Recommended Resources

Text Books

1. Cliff Ragsdale, 2018. Spreadsheet modelling and decision analysis: A practical introduction to business analytics, 8th ed., ISBN: 9789353502225, Cengage India.
2. Kumar, U. D., 2020. Business analytics: The science of data-driven decision making, 2nd ed., Wiley India.

Reference Books

1. Shmueli, G., Bruce, P. C., Yahav, I., Patel, N. R., & Lichtendahl, K. C., Jr., 2017. Data mining for business analytics: Concepts, techniques, and applications in R, Wiley.
2. Ryan, M., & Massaron, L., 2025. Advanced analytics for business: Generative AI and machine learning for tabular data, Manning Publications.

Further Readings

1. Frontline Systems, Inc. (n.d.). Analytic solver platform for education: User guide.
2. Frontline Systems. https://www.solver.com .

Suggested Courses

1. Predictive Modeling and Analytics https://www.coursera.org/lecture/predictive-modeling-analytics/7-how-to-build-a-model-using-xlminer-fM9fY
2. Best Logistic Regression Courses & Certificates [2026] https://www.coursera.org/lecture/data-analytics-business-capstone/building-logistic-regression-models-using-xlminer-vl92D

3. Predictive Modeling and Analytics https://www.coursera.org/lecture/predictive-modeling-analytics/7-building-neural-networks-using-xlminer-3eQXt
4. Best Cluster Analysis Courses & Certificates [2026] https://www.coursera.org/lecture/business-analytics-decision-making/5-cluster-analysis-with-xlminer-E53Hq

Semester/Year:	III Sem / II Year
Course Code:	26MBAB003

Title Of the Course:	Data Visualization for Decision Making
L: T: P: C:	3: 0: 2: 4

Overview

In this course, students will learn the fundamentals of data visualization and how to transform data into powerful visualizations that lead to actionable insights. Tableau, a leading business intelligence tool, will be used for hands-on learning. Students will create and share interactive dashboards that illustrate trends, variations, and data distributions through graphs and charts. Tableau can connect to files, relational databases, and big data sources to acquire and process information. The software also supports data blending and real-time collaboration, making it a unique and versatile tool. It is widely used by businesses, academic researchers, and government organizations for visual data analysis and is recognized as a leader in business intelligence and analytics by the Gartner Magic Quadrant. **Prerequisite:** Tableau Public software.

Course Objectives

1. To enable students with tools and techniques to communicate analytical insights
2. To empower students with understanding of appropriate visualization for the task in hand
3. To empower students with data blending options across data sources and bring them to life using visualizations
4. To enable them to create impactful dashboards for the problem in hand
5. To enable students to do storytelling for non-technical audience using data visualization

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts of data visualization and Tableau course, including interpreting analytical insights through appropriate visualizations.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of data visualization and Tableau in real-time problem solving by choosing appropriate visualizations for specific tasks.
3. Demonstrate the ability of analyzing real-time problems using the skills acquired from concepts, theories, models and principles of data visualization and Tableau by examining data sources and bringing them to life through visualizations.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of data visualization and Tableau for selecting impactful dashboards for given problems.
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of data visualization and Tableau, including developing analytical insights using Tableau story.

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case analysis collaboratively with student involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of data visualization by way of: ● Mini projects ● Submission of assignments ● Group assignments ● Written exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	Data Visualization Principles Data visualization principles, univariate categorical data, univariate numerical data, 1 categorical data & 1 numerical data, bivariate categorical data, bivariate numerical data, data-ink ratio and chart simplicity, Gestalt principles in data visualization, common pitfalls in axis scaling.	8
Unit II	Introduction to Tableau Introduction to Tableau, installing Tableau public, getting started with Tableau, saving Tableau public, navigation design flow, file types, data types, show me, terminology, dimensions vs measures, changing data types, filtering basics in tableau.	8
Unit III	Data Sources in Tableau Data sources, fields operation: Adding, combining, searching, reordering, editing metadata, data joining, data blending, connecting to google sheets, connecting to PDFs, pivoting data in Tableau, using data interpreter, extracting vs live data sources.	8
Unit IV	Tableau Features Tableau features, drill down, swapping dimensions, calculations, sort & filters, Tableau parameter, action URL, actions filter, quick table calculations, creating hierarchies, dynamic sorting.	8
Unit V	Storytelling with Tableau Storytelling with Tableau, Tableau charts: Bar, line, pie, crosstab, scatterplot, bubble, boxplot, histogram, tree map, dashboards, story, formatting, forecasting, trend lines, storytelling guidelines, highlight tables, dual-axis charts, KPI dashboards, interactive story points, exporting and publishing stories.	8

CO- PO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4
CO1	3	-	-	3	-	2	2	2	2
CO2	3	-	-	-	2	2	3	2	2
CO3	3	-	-	-	2	2	2	2	3
CO4	3	-	-	-	2	2	2	2	2
CO5	3	-	2	3	2	2	2	3	2

Action Based Component

1. Work with multiple datasets and data types.
2. Create Tableau charts on multiple datasets.
3. Create dashboards on multiple datasets.
4. Create story on multiple datasets.

Course Assessment

S.No.	Description of Assessment Method	Weightage %	Learning Outcomes Assessed				
			1	2	3	4	5
1	Assignment-1	15	X	X	X	X	X
2	Assignment- 2	15	X	X	X	X	X
3	Internal Assessment-1	15	X	X	X	X	X
4	Internal Assessment-2	15	X	X	X	X	X
5	Semester End Examination	40	X	X	X	X	X

Recommended Resources

Text Books

1. Loth, A., Vogel, N., & Sparkes, S., 2025. Visual analytics with Tableau: An Indian adaptation, 1st Indian ed., Wiley India.
2. Jones, B., 2014. Communicating Data with Tableau: Designing, developing, and delivering data visualizations, O'Reilly Media.

Reference Books

1. Knafllic, C. N., 2015. Storytelling with Data: A Data Visualization Guide for Business Professionals, John Wiley & Sons.
2. Monsey, M., & Sochan, P., 2020. Tableau for Dummies, Indian ed., Wiley India.

Further Readings

1. https://www.tutorialspoint.com/tableau/index.htm
2. https://public.tableau.com/en-us/s/resources
3. https://www.tableau.com/learn/articles/free-public-data-sets

Suggested Courses

1. Data Visualization with Tableau UCDavis, University of California https://www.coursera.org/specializations/data-visualization?
2. Data Management and Visualization — Wesleyan University https://www.coursera.org/learn/data-visualization

SEMESTER/YEAR:	III Sem / II Year
COURSE CODE:	26MBAB004
TITLE OF THE COURSE:	Predictive Analytics Using R
L: T: P: C:	3: 0: 2: 4

Overview

In this course, students will learn the basics of machine learning using R programming. They will gain an understanding of where machine learning fits within the analytics landscape, the different types of machine learning algorithms, and the overall data science process from data collection to model evaluation. The course is hands-on, using R and R Studio, with mini-projects implemented for each type of algorithm to reinforce practical learning.

Course Objectives

1. To understand and apply fundamental R data structures for data analysis
2. To explore the principles of exploratory data analysis and evaluate its significance in deriving insights from data
3. To design and implement supervised machine learning algorithms using R for business applications
4. To design and implement unsupervised machine learning algorithms using R for business application
5. To analyze and model time series data using R for forecasting and decision-making

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of machine learning course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of machine learning in real-time problem-solving by performing EDA and machine learning processes using R.
3. Demonstrate the ability of analyzing real-time problems using the skills acquired from concepts, theories, models and principles of machine learning by evaluating model performance with appropriate measures.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of machine learning for real-time problem-

solving and model appraisal.
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of machine learning, including choosing appropriate ML algorithms for different situations.

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case analysis collaboratively with student involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of R by way of: ● Mini projects ● Submission of assignments ● Group assignments ● Written exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	Introduction to Machine Learning (ML) Introduction to ML, types of analytics: Descriptive, diagnostic, predictive, prescriptive, statistics basics, types of data: Structured, semi structured, unstructured, structured data using RDBMS: Table, columns, rows, Introduction to ML, ML use cases, algorithm types: Supervised, unsupervised, reinforced, regression vs. classification, managing and understanding data: Vectors, factors, lists, data frames, matrices and arrays, importing and saving data from CSV files, exploring and understanding data: Exploring the structure of data, exploring numeric variables, exploring categorical variables, exploring relationships between variables, working with missing data in R, filtering and selecting data, summarizing data for business insights, exporting cleaned data to CSV or excel for business reporting.	8
Unit II	Supervised Machine Learning Supervised machine learning, KNN algorithm: How does it work?, example - Diagnosing breast cancer with KNN, Decision trees, Gini Index, chi square to choose decision node, parameters to control overfitting, example – Identifying risky bank loans, Neural networks: How do they work?, example - Modelling the strength of concrete with ANNs, multiple linear regressions, example – Predicting medical expenses using linear regression, logistic regression: Binary classification for business decisions, random forest algorithm: Concept and advantages over decision trees.	8
Unit III	Unsupervised Machine Learning Unsupervised machine learning, market basket using Apriori algorithm, support, confidence, lift, example – Identifying frequently purchased groceries with association rules, k means algorithm, example - Finding teen market segments using k-means clustering, hierarchical clustering - Visual method using dendrograms, business use: Grouping stores, regions, products.	8

Unit IV	Evaluating Model Performance Evaluating model performance, measuring performance for classification, working with classification prediction data, using confusion matrix to measure performance, measures other than accuracy: Kappa, sensitivity, specificity, precision and recall, F ratio, ROC, estimating future performance: Holdout, cross validation, bootstrap sampling, business interpretation of evaluation metrics, cost-based evaluation of classification errors.	8
Unit V	Time Series Introduction Time series introduction, data types and formats, components and objectives, reading, plotting and decomposing time series data in R, forecasts using exponential smoothing in R, ARIMA models in R, handling missing or irregular time series data, model evaluation for time series forecasting.	8

CO- PO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	2	2	2	2	2	2
CO2	3	2	3	2	2	2	3	2	2
CO3	3	2	2	1	2	2	2	2	3
CO4	3	2	2	1	2	2	2	2	2
CO5	3	2	2	1	2	2	2	3	2

Action Based Component

1. Exploratory data analysis using R
2. Supervised ML (KNN)
3. Unsupervised ML (Clustering)
4. Model evaluation using R
5. Time series using excel

Course Assessment

S.No.	Description of Assessment Method	Weightage %	Learning Outcomes Assessed				
			1	2	3	4	5
1	Assignment-1	15	X	X	X	X	X
2	Assignment- 2	15	X	X	X	X	X
3	Internal Assessment-1	15	X	X	X	X	X
4	Internal Assessment-2	15	X	X	X	X	X
5	Semester End Examination	40	X	X	X	X	X

Recommended Resources

Text Books

1. Forte, R. M., 2015. Mastering predictive analytics with R, Birmingham, UK: Packt Publishing.
2. Mayor, E., 2015. Learning predictive analytics with R, Packt Publishing Ltd.

Reference Books

1. Larose, D. T., 2015. Data mining and predictive analytics, John Wiley & Sons.
2. Tamhane, A. C., 2020. Predictive analytics: Parametric models for regression and classification using R, John Wiley & Sons.
3. Ledolter, J., 2013. Data mining and business analytics with R, John Wiley & Sons.

Further Readings

1. Wickham, H., & Golemund, G., 2017. R for data science: Import, tidy, transform, visualize, and model data, O'Reilly Media.
2. Matloff, N., 2011. The art of R programming: A tour of statistical software design, No Starch Press.
3. Golemund, G., 2014. Hands-on programming with R: Write your own functions and simulations, O'Reilly Media.

Suggested Courses

1. EDA <https://moderndive.netlify.app/1-getting-started.html>
2. <https://supervised-ml-course.netlify.app/> (4 Case Studies by Julia Silge, DataScientist from RStudio)
3. Dataiku Academy: Intro to Machine Learning
4. Dataiku Academy: Machine Learning Basics
5. Dataiku Academy: Scoring Basics
6. Dataiku Academy: Time Series Basics
7. Dataiku Academy Use Cases: Predictive Maintenance, Churn Prediction, Web Logs Analysis, Network Optimization, Bike Sharing Usage Patterns

Semester/Year:	IV Sem/ II Year
Course Code:	26MBAB005
Title Of the Course:	EDA Using Python
L: T: P: C:	3: 0: 2: 4

Overview

The programming requirements of data science demand a versatile and flexible language that is simple to code yet capable of handling highly complex mathematical processing. Python is particularly well-suited for these requirements, as it has established itself as a language for both general-purpose and scientific computing. Moreover, it is continuously upgraded with new additions to its extensive library ecosystem, catering to diverse programming needs. This is a hands-on practical course. Lab computers should preferably have Anaconda (Python IDE) installed, along with Jupyter Notebook, and a working internet connection. Students are not expected to have prior programming experience, but they must be willing to learn and practice beyond classroom sessions.

Course Objectives

1. To enable students, understand the features of Python language and different IDEs like Jupyter notebook and Spyder
2. To enable students, understand the features and functions of Python packages: NumPy, Pandas
3. To enable students to apply and leverage on Pandas and NumPy functionalities in performing data handling, data manipulation and data management tasks
4. To enable students to apply and leverage on Matplotlib library to perform datavisualizations and descriptive data analysis
5. To enable students to understand and apply the machine learning models using Python language

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of Python programming course, including its features and applications.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of Python programming in real-time problem solving by using NumPy for data handling and manipulation, and understanding the capabilities of the Pandas library.
3. Demonstrate the ability of analyzing real-time problems using the skills acquired from concepts, theories, models and principles of Python programming by examining NumPy and Pandas libraries for advanced data management tasks along with databases.

4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of Python programming for descriptive statistics and data visualization tasks.
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of Python programming, including adapting and applying different machine learning algorithms.

Approach to Learning	• Lectures • Active student participation and classroom exercises • Mini project • Case analysis collaboratively with student involvement • Visual presentation
Assessment Strategy	Participants will be assessed on both conceptual understanding and business applications of Python programming language by way of: • Case study discussions & solving mini projects • Submission of assignments and individual presentation • Written exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	Introduction to the Data Analysis and Python World Introduction to the data analysis and Python world, data analysis process, quantitative and qualitative data analysis, Python introduction, Python interpreter, Python distributions, Anaconda environment, Jupyter notebook, aide for Python, built-in Python libraries, built-in functions, data types, libraries and modules, programs on applications of data types methods and built-in functions, introduction to google Collab as an aide, best practices for clean business code in Python, basics of error handling.	8
Unit II	Introduction to NumPy and Pandas Libraries , NumPy: Creation of an array, operators, functions, indexing, slicing, and iterating, shape manipulation, array manipulation, structured arrays, reading and writing array data on files, Pandas: Introduction to Pandas data structures, reindexing, dropping, arithmetic and data alignment, operations between data structures, function application and mapping, sorting and ranking, “not a number” data, hierarchical indexing and levelling, creating custom functions, data filtering and conditional selection, exporting data to excel and CSV formats.	8

Unit III	Data Handling with Pandas Data handling with Pandas, reading data in CSV or text files, reading and writing HTML files, reading data from XML, JSON data, Pickle—Python object serialization, reading and writing data on Microsoft excel files, interacting with databases, data preparation, concatenating, data transformation, discretization and binning, permutation, string manipulation, data aggregation, group iteration, advanced data aggregation, data profiling and summary reports, detecting and handling duplicates, data validation techniques in Pandas.	8
Unit IV	Descriptive Statistics and Data Visualization Descriptive statistics and data visualization, descriptive statistics: Distribution functions, measures of central tendencies, measures of variations, measures of shape, visualizations: Visualization with Pandas library, visualization with Matplotlib library, choosing the right chart for the right data, outlier detection through visualization.	8
Unit V	Model Development and Evaluation Model development and evaluation, hypothesis testing, P-hacking, correlation and regression, linear vs. non-linear models in simple terms, constructing and implementing linear models, descriptive, supervised machine learning, unsupervised machine learning, reinforcement learning, Statistics: Unified machine learning workflow, exploratory data analysis on a sample dataset, data splitting: Train/test vs. cross-validation, model evaluation metrics (Accuracy, precision, recall, F1-score).	8

CO- PO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	2	2	2	2	2	2
CO2	3	2	2	2	2	2	3	2	2
CO3	3	2	2	2	2	2	2	2	3
CO4	3	2	3	2	2	2	2	2	2
CO5	3	2	2	2	2	2	2	3	2

Action Based Component

1. Mini projects on data collection, data cleaning, data transformation, and data visualization, followed by exploratory data analysis and interpretation.
2. Group-based presentations and reports on exploratory data analysis outcomes, emphasizing business interpretation and managerial insights.

3. Data preprocessing and validation activities, including handling missing values, outliers, and duplicate records in real datasets.

Course Assessment

S.No.	Description of Assessment Method	Weightage %	Learning Outcomes Assessed				
			1	2	3	4	5
1	Assignment-1	15	X	X	X	X	X
2	Assignment- 2	15	X	X	X	X	X
3	Internal Assessment-1	15	X	X	X	X	X
4	Internal Assessment-2	15	X	X	X	X	X
5	Semester End Examination	40	X	X	X	X	X

Recommended Resources

Text Books

1. Nelli, F., 2018. Python data analytics with Pandas, NumPy, and Matplotlib, Apress.
2. Mukhiya, S. K., & Ahmed, U., 2020. Hands-on exploratory data analysis with Python, Packt Publishing.

Reference Books

1. Jason S. Schwarz, Chris Chapman & Elea McDonnell Feit, 2020. Python for Marketing Research and Analytics, Springer International Publishing.
2. Wes McKinney, 2018. Python for Data Analysis, O'Reilly Media.
3. Joel Grus, 2019. Data Science from Scratch, O'Reilly Media.

Further Readings

1. Oluleye, A., 2023. Exploratory Data Analysis with Python Cookbook: Over 50 recipes to analyze, visualize, and extract insights from structured and unstructured data, Packt Publishing Ltd.
2. Jake VanderPlas, 2016. Python Data Science Handbook, O'Reilly Media.

Suggested Courses

1. University of Michigan. (n.d.). Python for Everybody Specialization. Coursera. <https://www.coursera.org/specializations/python>
2. University of Michigan. (n.d.). Crash Course on Python. Coursera. <https://www.coursera.org/learn/python-crash-course>
3. University of Michigan. (n.d.). Programming for Everybody (Getting Started with Python). Coursera. <https://www.coursera.org/learn/python>